

COLOUR DECONVOLUTION

Similarly to the command *Image ► Color ► RGB split*, this plugin unmixes an RGB image produced by subtractive mixing (inks, histological dyes) into separate channels corresponding to up to 3 determined colors. This is useful e.g. to do pigment separation.

Documentation

See [Gabriel Landini](#) webpage.

This  implements the method described in the following paper:

Ruifrok, A.C. & Johnston, D.A. (2001), "Quantification of histochemical staining by color deconvolution", *Anal. Quant. Cytol. Histol.* **23**: 291-299, PMID 11531144

Version history

- 30/Mar/2004 released - based on macro code from A.C. Ruifrok.
- 03/Apr/2004 resolved ROI exiting
- 07/Apr/2004 added Methyl Green DAB vectors
- 08/Jul/2004 shortened the code
- 01/Aug/2005 added fast red/blue/DAB vectors
- 02/Nov/2005 changed code to work with image stacks (DLC - dchao at fhcrc org)
- 02/Nov/2005 changed field names so user-defined colours can be set within macros (DLC - dchao at fhcrc org)
- 04/Feb/2007 1.3 disable popup menu when right clicking
- 23/May/2009 added Feulgen-light green vectors
- 14/Apr/2010 v 1.4 added Giemsa vector (Methylene blue & eosin). The images are now names "title"-(Colour_1) etc so there are not clash of names when using []. The log window now prints the java code of the translation matrix to include new vectors in the plugin. Added "Hide legend" option.
- 22/Jun/2010 v 1.5 added Masson Trichrome vector (Methyl blue & Ponceau Fuchsin only (this does not have Iron Haematoxylin vector!). Fixed bug: check for 0 components before hiding legend (otherwise there was no image shown if legend hidden)
- 26/Mar/2011 v.1.6 added Brilliant_Blue stain.
- 03/Aug/2011 v1.7 added progress bar (thanks to Oskari Jaaskelainen), added warning about immunostains.
- 17/Oct/2015 v 1.8 B. Pavie refactorized the code into multiple classes and made some public method to generate the result ImageStacks without displaying them so it can be called from scripts.

Menu path

Image ► Color ► Colour Deconvolution

Retrieved from "https://imagej.net/index.php?title=Colour_Deconvolution&oldid=36340"

This page was last modified on 3 October 2017, at 14:29.

[Privacy policy](#) [About ImageJ](#) [Imprint](#)

Categories: [Plugins](#) | [Color processing](#)

Colour Deconvolution	
Project	Fiji
URL	https://imagej.net/Colour_Deconvolution
Source	on GitHub
License	GPLv3
Release	3.0.2
Date	Sat Apr 29 22:36:16 CDT 2017
Development status	Active
Support status	Active
TEAM	
Founders	Gabriel Landini
Leads	Gabriel Landini
Developers	Gabriel Landini
Debuggers	Gabriel Landini
Reviewers	Gabriel Landini
Support	Gabriel Landini
Maintainers	Gabriel Landini, Curtis Rueden
Contributors	Johannes Schindelin, Mark Hiner, Benjamin Pavie